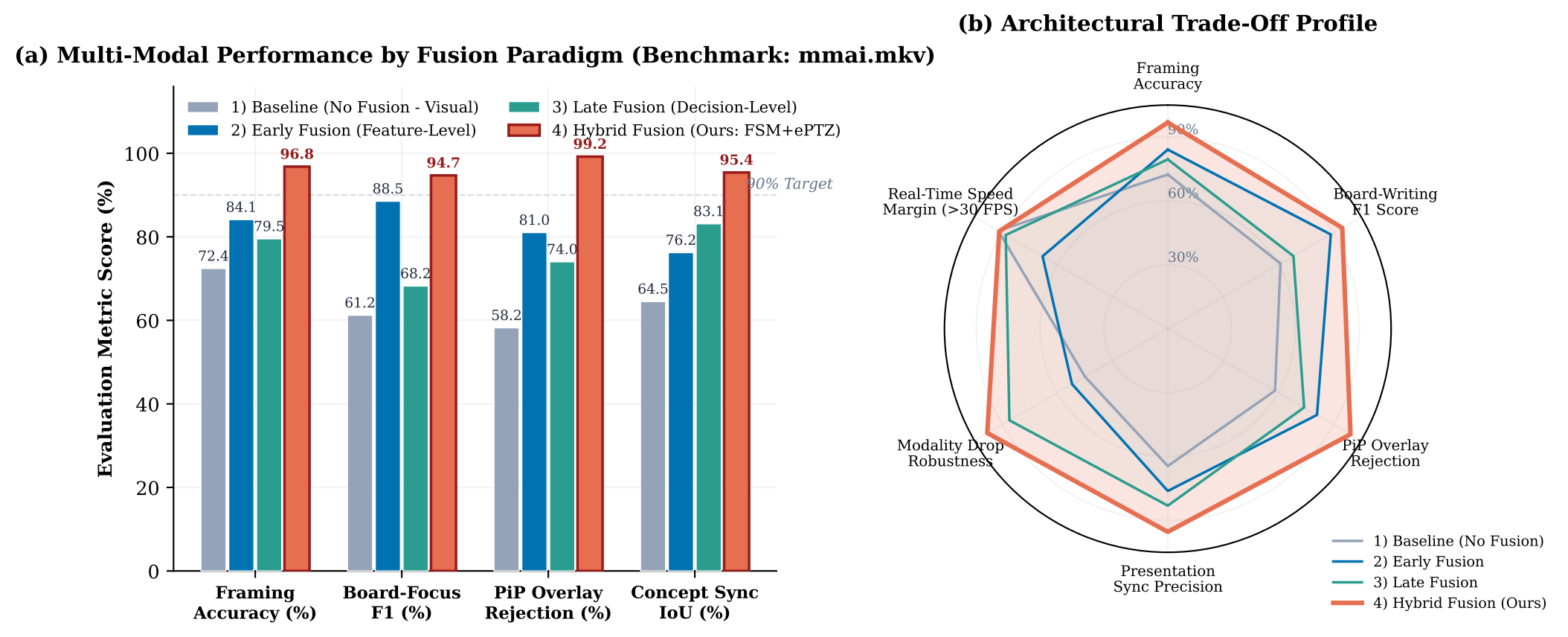
(a) Multi-Modal Performance by Fusion Paradigm (Benchmark: mmai.mkv)
1) Baseline (No Fusion - Visual)
2) Early Fusion (Feature-Level)
3) Late Fusion (Decision-Level)
4) Hybrid Fusion (Ours: FSM+ePTZ)
Evaluation Metric Score (%)
90% Target
72.4
84.1
79.5
96.8
61.2
88.5
68.2
94.7
58.2
81.0
74.0
99.2
64.5
76.2
83.1
95.4
Framing Accuracy (%)
Board-Focus F1 (%)
PiP Overlay Rejection (%)
Concept Sync IoU (%)
(b) Architectural Trade-Off Profile
Framing Accuracy
Board-Writing F1 Score
PiP Overlay Rejection
Presentation Sync Precision
Modality Drop Robustness
Real-Time Speed Margin (>30 FPS)
90%
60%
30%
1) Baseline (No Fusion)
2) Early Fusion
3) Late Fusion
4) Hybrid Fusion (Ours)